

Joint Control Barrier Function for Target Visibility and Thrust Deliverability

A camera-plane QP over the commanded inertial acceleration

Soft–Precise–Landing

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Scope and notation. This note follows the notation of the manuscript’s Visibility CBF subsection verbatim (frames $\mathcal{I}/\mathcal{C}$, feature points $\tilde{\mathbf{r}}_i$, coupling $\tilde{L}_{\omega,i} = L_{\omega}M$, lean $\boldsymbol{\theta}$, cap θ_{cap} , drift $\mathbf{d}_i$, look-ahead τ) and extends it with one additional symbol, A_{cap} , the vehicle’s maximum achievable thrust-to-mass ratio. Visibility is fundamentally an *attitude* constraint (the body tilt shifts the FoV), but the attitude actually commanded is a byproduct of the desired inertial acceleration ${}^{\mathcal{I}}\mathbf{a}_d$. Rather than solving the tilt-only projection of the manuscript and separately deciding how much vertical thrust to spend, this note poses one projection directly over ${}^{\mathcal{I}}\mathbf{a}_d$ that jointly enforces (i) the barrier condition of the manuscript’s Visibility Forward Invariance theorem and (ii) the deliverable-lateral-acceleration bound of the manuscript’s Property 1, as a single constrained problem rather than a tilt projection followed by a separate deliverability cap. The coupling itself is unchanged: the **rotational part of the IBVS interaction matrix**, L_{ω} , evaluated at the measured image point — exact, *depth-independent*, no de-rotation or virtual frame required.

Notation

Symbol	Meaning
$\tilde{\mathbf{r}}_i, \tilde{r}_{i,k}$	measured camera-frame feature point i , tangent units, per axis k
$L_\omega(\tilde{\mathbf{r}}_i)$	rotational interaction matrix (roll/pitch columns), $\partial \dot{\tilde{\mathbf{r}}}_i / \partial \boldsymbol{\omega}_{\text{rp}}$
$\boldsymbol{\omega}_{\text{rp}} = [\omega_\phi, \omega_\theta]$	body roll/pitch rate (the <i>rotation-axis</i> vector)
$M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	lean→rotation map: $\boldsymbol{\omega} = M\boldsymbol{\theta}$ (lean $+x$ is a rotation about $+y$)
$\tilde{L}_{\omega,i} = L_\omega M$	effective tilt coupling (orthogonal M , so row norms are preserved)
$\mathcal{I}\mathbf{a}_{\text{d}}$	desired inertial acceleration; this projection’s decision variable
$\mathcal{I}a_{\text{d},z}$	desired vertical acceleration component ($a_z \equiv \mathcal{I}a_{\text{d},z} $)
$\boldsymbol{\theta} = R_z(-\psi_{\text{b}})(\mathcal{I}\mathbf{a}_{\text{d},xy}/\mathcal{I}a_{\text{d},z})$	<i>derived</i> lean-direction vector
$\boldsymbol{\theta}_{\text{curr}}$	current tilt, from the realized attitude
A_{cap}	vehicle’s maximum achievable thrust-to-mass ratio
θ_{cap}	deliverable-tilt cap (60° nominal; a 2×-hover-thrust margin)
$\mathbf{d}_i$	exogenous (translation) feature drift, $\mathbf{d}_i = \dot{\tilde{\mathbf{r}}}_{i,\text{obs}} - L_\omega \boldsymbol{\omega}_{\text{rp}}^{\text{curr}}$
$\mathcal{R}, f$	resolution, focal (px); $\varphi_{\text{max}} = \mathcal{R}/(2f)$ half-FoV (tangent), the FoV edge
τ, g	drift look-ahead; gravitational acceleration

1 Barrier on the real camera plane

Per image axis k , for feature point i ,

$$h_{i,k} = \varphi_{\text{max},k} - |\tilde{r}_{i,k}|. \quad (1)$$

$h_{i,k} \geq 0 \iff$ feature point i is on the sensor on axis k . $\tilde{\mathbf{r}}_i$ is the *measured* camera-frame feature point — nothing is de-rotated. This is the manuscript’s own barrier, unchanged.

2 Tilt→feature coupling, expressed in acceleration units

A camera rotation moves a feature point by the rotational part of the IBVS interaction matrix, a standard and **depth-independent** object evaluated at the measured point:

$$\dot{\tilde{\mathbf{r}}}_i = L_\omega(\tilde{\mathbf{r}}_i) \boldsymbol{\omega}_{\text{rp}} + \mathbf{d}_i, \quad L_\omega = \begin{bmatrix} xy & -(1+x^2) \\ 1+y^2 & -xy \end{bmatrix}_{(x,y)=\tilde{\mathbf{r}}_i}. \quad (2)$$

L_ω contains **no depth**. The drift is measured directly by stripping the rotational term from the observed feature velocity, $\mathbf{d}_i = \dot{\tilde{\mathbf{r}}}_{i,\text{obs}} - L_\omega \boldsymbol{\omega}_{\text{rp}}^{\text{curr}}$ (using the true gyro rate), so it carries only genuine target motion, not our own tilting.

A commanded lean produces a rotation about the perpendicular axis, $\boldsymbol{\omega} = M\boldsymbol{\theta}$, and the lean is itself the direction of the commanded lateral acceleration relative to the commanded vertical

acceleration, $\boldsymbol{\theta} = R_z(-\psi_b)({}^{\mathcal{I}}\mathbf{a}_{d,xy}/{}^{\mathcal{I}}a_{d,z})$ — the manuscript’s own definition. Substituting both relations into (2) gives the feature velocity directly as a function of the acceleration command:

$$\boldsymbol{\omega} = M\boldsymbol{\theta}, \quad \tilde{L}_{\omega,i}\boldsymbol{\theta} = \underbrace{\left(\frac{\tilde{L}_{\omega,i} R_z(-\psi_b)}{a_z}\right)}_{N_i(a_z)} {}^{\mathcal{I}}\mathbf{a}_{d,xy}. \quad (3)$$

$N_i(a_z) = \tilde{L}_{\omega,i} R_z(-\psi_b)/a_z$ is the barrier’s Jacobian with respect to the lateral acceleration directly. It depends on a_z — the same quantity the joint projection below also solves for — which is exactly the coupling that projection is built to resolve, rather than treating a_z as an external given. For a finite acceleration change the exact map is the rotation homography of the measured point; the first-order form

$$\tilde{\mathbf{r}}_i({}^{\mathcal{I}}\mathbf{a}_{d,xy}) \approx \tilde{\mathbf{r}}_i + N_i(a_z){}^{\mathcal{I}}\mathbf{a}_{d,xy} - \tilde{L}_{\omega,i}\boldsymbol{\theta}_{\text{curr}} \quad (4)$$

is the local linearization, accurate over one control step, exactly matching the manuscript’s own linearization of $\dot{h}_{i,k}$ along (2). $\boldsymbol{\theta}_{\text{curr}}$ is read directly from the realized attitude (not from ${}^{\mathcal{I}}\mathbf{a}_d$), so it needs no rotation from acceleration units.

3 The visibility–deliverability projection

Two physical constraints bind the same variable ${}^{\mathcal{I}}\mathbf{a}_d$: the barrier condition of (1) along the feature response (2)–(3), and the deliverable-lateral-acceleration bound of the manuscript’s Property 1 (inner-loop deliverable-tilt saturation bounds $\|{}^{\mathcal{I}}\mathbf{a}_{d,xy}\|$ by a state-dependent fraction of $|{}^{\mathcal{I}}a_{d,z}|$). Writing ${}^{\mathcal{I}}\mathbf{a}_d^*$ for the conditioned output (paralleling the manuscript’s $\boldsymbol{\theta}^*$ notation for its own tilt-only projection),

$$\boxed{\begin{aligned} & {}^{\mathcal{I}}\mathbf{a}_d^* = \arg \min_{{}^{\mathcal{I}}\mathbf{a}_d} \|{}^{\mathcal{I}}\mathbf{a}_d - {}^{\mathcal{I}}\mathbf{a}_d^0\|^2 \\ \text{s.t. } & |\tilde{\mathbf{r}}_i + N_i(a_z){}^{\mathcal{I}}\mathbf{a}_{d,xy} - \tilde{L}_{\omega,i}\boldsymbol{\theta}_{\text{curr}} + \tau \mathbf{d}_i| \preceq \boldsymbol{\varphi}_{\text{max}}, \quad \|{}^{\mathcal{I}}\mathbf{a}_d + g\mathbf{e}_3\| \leq A_{\text{cap}} \end{aligned}} \quad (5)$$

where ${}^{\mathcal{I}}\mathbf{a}_d^0$ is the unconstrained desired acceleration — the manuscript’s own ${}^{\mathcal{I}}\mathbf{a}_d$ before conditioning, the target this projection stays closest to. The first constraint is the manuscript’s per-feature FoV box, now written directly on ${}^{\mathcal{I}}\mathbf{a}_{d,xy}$; the second is the true thrust-deliverability sphere in gravity-shifted acceleration space (the vehicle’s total thrust vector is $m({}^{\mathcal{I}}\mathbf{a}_d + g\mathbf{e}_3)$, and $\|{}^{\mathcal{I}}\mathbf{a}_d + g\mathbf{e}_3\| \leq A_{\text{cap}}$ is exactly the statement that this stays within the vehicle’s maximum). Both are convex for a fixed a_z ; since $N_i(a_z)$ itself depends on a_z , the two constraints are resolved jointly rather than sequentially: the lateral acceleration satisfying the box and the vertical acceleration satisfying the sphere are solved for together, not one after the other. ${}^{\mathcal{I}}\mathbf{a}_d^*$ then defines the desired attitude and thrust magnitude through the inner loop, exactly as the manuscript’s own $\boldsymbol{\theta}^*$ does.

- **Directional** — only the outward-binding box row is active; inward (recovery) and tangential accelerations are free. No strangling, matching the manuscript’s own $\dot{h}_{i,k}$ argument.
- **Envelope-aware** — the deliverability side of (5) is the vehicle’s actual maximum thrust-to-mass ratio A_{cap} , not θ_{cap} applied *as if* a_z were fixed at hover; it tightens or loosens correctly as a_z moves away from hover.
- **Forward-invariant** — the look-ahead shifts the box anchor by $\tau \mathbf{d}_i$, so the projection accounts for centroid drift over the next τ s, exactly as in the manuscript.

- **Bounded by construction** — because $\mathcal{I}\mathbf{a}_d^*$ always satisfies the sphere constraint, the derived lean $\boldsymbol{\theta}^* = R_z(-\psi_b)\mathcal{I}\mathbf{a}_{d,xy}^*/a_z^*$ satisfies $\|\boldsymbol{\theta}^*\| \leq \tan \theta_{\text{cap}}$ at the solved a_z^* automatically; no separate, hover-assumed cap application is needed.

4 Feasibility

The box constraint alone is **infeasible** iff $\varphi_{\max,k} < 0$ for some k (the marker subtends the full FoV). The sphere constraint is always feasible by construction (it is satisfied by $\mathcal{I}\mathbf{a}_d = \mathbf{0}$). Genuine joint infeasibility — the box demanding more lateral acceleration than A_{cap} can deliver at any achievable a_z — is the acceleration-space counterpart of the manuscript’s feasibility condition

$$\sigma_{\min}(\tilde{L}_{\omega,i})(\tan \theta_{\text{cap}} - \|\boldsymbol{\theta}_{\text{curr}}\|) \geq \max_{i,k} (|\tilde{r}_{i,k}| - \varphi_{\max,k})_+ + \bar{d} \quad \forall i, \quad (6)$$

evaluated here at the solved a_z^* rather than at a fixed hover-assumed θ_{cap} : joint feasibility holds whenever some achievable a_z satisfies (6), a weaker requirement than demanding it hold at the hover value of a_z alone.

5 Assumptions

1. **The coupling $\tilde{L}_{\omega,i} = L_\omega M$ and its sign convention are calibrated.** The fixed 90° map $\boldsymbol{\omega} = M\boldsymbol{\theta}$ bridges the lean direction and the rotation-axis rate that L_ω couples to.
2. **The conditioned acceleration is realized:** the inner attitude loop tracks the attitude built from $\mathcal{I}\mathbf{a}_d^*$ on a timescale fast relative to the outer loop, the manuscript’s own timescale-separation assumption.
3. **The drift d_i is the translation part** of the observed feature velocity, $\dot{\tilde{\mathbf{r}}}_{i,\text{obs}} - L_\omega \boldsymbol{\omega}^{\text{curr}}$, entering as *measured* motion, matching the manuscript’s own treatment — no separate disturbance term is introduced.
4. **First-order L_ω over the cycle.** Exact is the rotation homography of $\tilde{\mathbf{r}}_i$; the linearization (4) is accurate over one control step, the same approximation the manuscript’s own barrier derivative relies on.
5. **A_{cap} is the vehicle’s true achievable thrust-to-mass ratio,** the physical quantity the manuscript’s own $\theta_{\text{cap}} = 60^\circ$ (a 2 $\times$ -hover-thrust margin) is derived from — this projection uses that quantity directly rather than the fixed angular margin computed from it at hover.

6 Scale-freeness and depth-freeness

The barrier (1), the coupling (2), and the projection (5) contain only the measured image point $\tilde{\mathbf{r}}_i$, $\varphi_{\max} = \mathcal{R}/(2f)$, the body rate, the commanded acceleration, and the vehicle’s own A_{cap} and g . L_ω **is depth-independent by construction**, and A_{cap} , g are fixed vehicle/physical constants, not measurements of target depth or altitude — the whole formulation remains manifestly scale- and depth-free, matching the manuscript’s own scale-free design throughout.